EED-56NAV-03

STATEMENT OF TECHNICAL REQUIREMENTS

FOR

ELECTROMAGNETIC LOG (COTS)

EED-56-03

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STATEMENT OF TECHNICAL REQUIREMENT (SOTR) FOR ELECTRO MAGNETIC LOG

1. INTRODUCTION.

Electromagnetic Log is an instrument used to measure the speed and distance traveled by the ship. The operation of the Electromagnetic log is based upon a physical phenomenon, i.e. The Electro Magnetic Induction. The system consists of a pick up head (Transducer) fitted at the Hull of the ship which when excited generates a magnetic field in the sea water. When the ship moves an Electromagnetic force proportional to the speed of the ship, is induced across the electrodes the detected signal from the transducer is fed to the log unit which generates a slow varying voltage proportional to the speed of the ship.

2. SPECIFICATION OF EM LOG

(a) Circuitry – Micro controller based.

(b) Speed Range - ± 60 knots

(c) Distance - 0 to 99999.9 NM (Re-settlable to 0)

(d) **Accuracy at Factory**

(i) Speed = 0.1 knots upto 10 knots & ± 1% above 10 knots

(ii) Features - Speed range - + 60 knots corrected to one decimal digit

(e) **Accuracy at Sea.**

Speed = ± 0.3 knots upto 10 knots
± 3% above 10 knots

(f) **Displays.**

Speed : 3 integer, 1 decimal 7 seg.red LED

Distance : 5 integer, 1 decimal 7 seg.red LED

Total distance : 6 integer 7 segment red LED

Unit : Single LED

(g) **Power** : 239 / 115V, single phase, 50/60Hz, 100VA

(h) **Outputs:**

(i) Digital speed and partial distance to RDU – RS 422
(ii) 400 pulses / nautical mile to RTU

(iii) Digital speed in RS 232C form interface to speed Retransmission. Unit

3. **ELECTRICAL**

(a) **MASTER UNIT:-** The master unit consists of the Power Supply Unit, Control Unit assembly, Front Panel assembly and Bottom Plate Assy which receives the speed signal from the transducer and compute the speed and distance information and following:-

(i) **Input to Master Unit:-** AC Power 230/115V, 50/60Hz,300VA (MAX) Signal input from the transducer 50 micro volts/knots approx.

(ii) **Out put of Master Unit**

(aa) **Display**
- Speed
- Partial distance
- Total distance

} LCD display

(ab) Speed and partial distance to Repeaters - RS485, serial

(ac) Speed information to RTU - RS 232, serial

(ad) Pulsed output to RTU - 400 Pulses/NM, serial

(ae) Power output to Distribution - 230/115V, 50/60 Hz, 200VA(MAX)

(iii) **Features**

(aa) Speed range - $\pm$ 60 knots corrected to one decimal digits

(ab) Speed accuracy - $\pm$ 0.1 knots up to 10 knots and $\pm$ 1% above 10 knots under factory test conditions $\pm$ 0.3 knots up to 10 knots and $\pm$ 3% above 10 knots installed on ships

(ac) Distance Accuracy - 0.5% on the calculated speed

(ad) Circuitry - Microcontroller based

(ae) Total distance change - 0 to 999999 NM, re settable through Micro Switch

(af)	Partial distance change	-	0 to 9999.9 NM, reset table through Micro Switch
(ag).	Calibration	-	8 points

4. **Mechanical**

(a) **Master Unit**

(i)	Construction	-	EMI protected sheet metal enclosure environmentally protected to IP 52
(ii)	Mounting	-	Mounted on Bulkhead with shock mounts
(iii)	Dimensions	-	(not more than) 1. Height: 500mm 2. Width : 410mm 3. Depth : 380mm
(iv)	Weight	-	$\leq$ 50 Kg

5. **Junction Box**. The Junction box is used to provide connection between the pick up head and the EM Log Master Unit when these are set far apart from one another. A sealing compound is used to seal the junction box against moisture

6. **Transducer:-** The Hull fitting consists of a 76.2 mm sea valve which is bolted to a plate welded to the hull. A locating flange is bolted to the top flange of the valve .The valve flange and locating flange are supplied un drilled to enable the ship builder to site the valve in the optimum position. The locating consists of two stainless steel swivel bolts which either clamp the transducer gland or if the transducer is removed a plug nut (blanking off plug). The approx specifications given below:

(a)	Type	-	Pattern No; Retractable / Fixed flush type
(b)	Make	-	Name of the OEM
(c)	Input	-	13 Vrms, 13.5 Hz constant current 0.599 Amps
(d)	Sensitivity	-	50 Micro volts /knot approx
(e)	Construction	-	Material Specifications, type of fitment (e.g Gun Metal casting with retractable transducer using sea valve)
(f)	Mounting	-	Detail installation drawing to be provided
(g)	Weight	-	Not more than 35 Kgs.

7. **Repeater Distribution Unit. (RDU)** This unit receives the speed, distance and unit information from Master Unit and distributes it to different repeaters using RS 422 line drives. This AC power required for the repeaters are also taken from the RDU along with data.

(a) **Input (from Master Unit)**

(i)	AC power	-	230/115V, 50/60Hz, 200VA (MAX)
(ii)	Speed and Distance	-	RS 485, Serial
(ii)	Speed	-	RS 232, Serial
(iv)	Distance as Pulses (for RTU)	-	RS 422, 400P/NM

(b) **Output**

(i)	AC power (for Repeaters)	-	18 ± 1 Vrms, 50/60 Hz
(ii)	Speed and Distance	-	RS 485, Serial
(iii)	Speed	-	RS 232, Serial
(iv)	Distance as Pulses (for RTU)	-	RS 422, 400P/NM

(c) Construction - Sheet metal enclosure with hinged door

(d) Mounting Bulkhead on shock mount

(e) Dimensions. - (not more than)
(i) Height: 400 mm
(ii) Width : 250 mm
(iii) Depth : 150 mm

(f) Weight - Not more than 10 Kg

8. **Repeater.** The repeaters get speed and partial distance data and AC power from RDU and display the speed and partial distance information of EM Log. A two wire transmission is used to send the data, clock and strobe for the display in the repeaters. A baud rate of 1200 is used to send the data. A low going signal marks the start of the frame. Next two cells contain one bit of data. After sending one bit, the line is raised to high and again brought to low for a start bit.

(a) **Input**

(i)	AC power	-	18 ± 1 Vrms, 50/60 Hz
(ii)	Speed and distance	-	RS 485, Serial

(b) **Output**

(i) **Display**

(aa) Speed	-	3 integer, 1decimal, 7 Seg. Red LED
(ab) Partial distance	-	5 integer, 1 decimal, 7 Seg Red LED

(c) Construction - (i) BH type
(ii) VCS type

(d) Mounting - Bulkhead/panel mounting

(e) Dimensions - a) BH type – 192mm x 192mm x 90mm
b) VCS type – 151.61mm X 151.61mm X 173 mm

(f) Weight - Not more than 3Kg

9. **Retransmission Unit.** This unit receives the two EM Log output viz the speed in RS232 format and the distance as 400 P/NM. Pulses & convert these information into different formats to suit the input format requirements of different equipments onboard the ship.

(a) **Input**

(i)	AC power	230/115V, 50/60Hz, 500VA (VA may change from ship to ship)
(ii)	Reference Supply	115V, 400Hz
(iii)	Speed	RS 232, Serial
(iv)	Distance	400 Pulses/NM, RS 422

(b) **Output**

(i)	Synchro / resolver	5 Nos.* (*Configurable depending
(ii)	Potential free contacts	10 Nos * on ship's requirement)

(c) **Accuracy** - 24 Arc Min

(d) Construction - Sheet metal enclosure with hinged door
(e) Mounting - Bulkhead on shock mount
(f) Dimensions. - Height: 425 mm
Width : 330 mm
Depth : 300 mm

(g) Weight - Not more than 10 Kg

10. **ENVIRONMENTAL SPECIFICATIONS**

(a)	Vibration	5 to 33 Hz frequency sweep for ½ hr. and Endurance at resonance for 1 hr. if no resonance, endurance at 3 spot frequencies for 1 hr. each on all three axes. Amplitude: $\pm$ 0.125 mm constant displacement (JSS 55555 Test No.28)
(b)	High Temperature	55 C $\pm$ 3 C for 16 hrs. Procedure 5, Test condition G (JSS 55555 Test No.17)
(c)	Damp Heat	Temp: 40 C $\pm$ 2 C, Relative Humidity: 95% for 16 hrs. Procedure 4(JSS 55555 test No.10)
(d)	Low temperature	0 for 16 hrs Procedure 4 (JSS 55555 Test No.20)
(e)	Drip Proof	Vertical water droplet 1m height for 13 mts. (JSS 55555 Test No.11)
(f)	Tropical Exposure	Introduce the equipment into the chamber maintained at a temperature of 20 °C $\pm$ 5° C. Raise the temp. and RH of the chamber to 35° C + 2° C and 95% respectively over a period of 3 hrs. Maintain the above condition for a Period of 12 hours. Lower the temperature to 20 °C $\pm$ 5° C over a period of 3 hours. Maintain this temperature for a of 6 hours. Testing condition C-28 cycles (JSS 55555 Test No.27).
(g)	Mould Growth	Testing will be confined to representative samples/components/materials (JSS 55555 Test No.21)
(h)	Corrosion Salt	Testing will be confined to representative samples/components/materials (JSS 55555 Test No.9)
(j)	Bump	4000 + 10 Bumps, 40g, 6 msec., 1 to 3 bumps/sec. (JSS 55555 Test No.5)
(k)	Shock or Impact	50g, 11msec or 30g, 18msec 3 shocks each of 6 faces. Procedure 2. Test method A – Basic design test (JSS 55555 Test No.24)
(m)	EMI/EMC	As per MIL 461C.

11. **GENERAL REQUIREMENT.**

(a) The system will be suitable for 240V, 1 Ph. 50 Hz AC ship's supply. This ship supply will be provided to the EM Log Master unit, RDU & RTU through 2 x 1.5 sq. mm cable of approx overall diameter of 10mm.

(b) All interconnection cables except power supply cable and output feed cable from the RTU will be supplied by

(c) All connectors including those for power supply cable will be supplied by M/s Keltron

(d) Approved HAT/SAT schedule indicating speed accuracy are to be supplied by OEM. Approval of HAT/SAT schedules from Naval Authority is OEM responsibility.

(e) RTU shall have provision of speed output in pulse (200 p/nm & 400 p/nm) format also. RTU shall have digital output feed to cater the interface requirement of ECDIS.

(f) To use the potential free pulse out put (200 /nm & 400 p/nm) of log RTU, the individual consumer side should have provision of providing 5-24V supply (max 3A)) to the Opto coupler already fitted in RTU of Log.

(g) No separate route is required for the EM Log system cables except Transducer cable upto the Master unit and can be laid along with power cable through the ship's existing cable route. Transducer Cable is to be laid in separate route, at least 100mm away from power cable.

(h) The system is immune to EMI/EMC problems. No additional precaution is required to be taken by the Yard for the same.

(j) Length of Pick-up cable and transducer cable from JB to Master unit can be cut to suit ship's requirement without affecting performance of the system. There shall be no length restrictions for other interconnection cables of the system excepting cables from Log Master Unit to RTU which will be restricted to 8 M each.

(k) All the installation drawings with fixing & mounting details, Interconnection cable diagram with connector /termination details will be supplied within four weeks from placement of order / LOI, Interconnection cable diagram should be complete with material list indicating type, part no against each item. Approval of Drawings from Naval Authority is OEM responsibility.

(m). All other points as discussed in the TNC are also applicable.

12. **Onboard Spares**. The onboard spares as per section 11 of NES 636 caters for one set of onboard spares recommended by the supplier and approved by NHQ shall be supplied along with the main equipment. Equipment PIL indicating item-wise cost is to be supplied along with the list of onboard spares while obtaining NHQ approval.

13. **Inspection**

(a) Inspecting Authority: DQA (N)
(b) Receipt Inspection: Respective WOTs.

14. **Service Engineer** - OEM will provide Service Engineer (15 days in 2 sorties) for supervision of installation, setting to work, attending Harbour Acceptance Trials (HAT) and Sea Acceptance Trials Trials (SAT) and commissioning of the equipment to the satisfaction of the Surveyor / Owner. All necessary commissioning spares, special tools test/calibration kit etc, as required shall be supplied free of cost and shall be brought by the Service Engineer during STW/commissioning of the system. Service Engineer, if required beyond the above specified time would be at extra charge.

16. **Documentation**

S.No.	Description	Qty
(a)	As made drawing	27 sets on reproducible /ship -27 sets paper print /ship -3 sets on CD/Ship
(b)	Interactive Electronic Technical Manuals(IETM) consisting of following Documents :- -User Hand Book - Instruction for Installation - Technical Manual Part I – Technical Information (Volume 1 & 2) Part II – Maintenance Part III – Manufacturer's Parts List (Vol.1&2)	3 sets on CD
(c)	HAT/SAT Schedules	-3 sets paper print /ship -3 sets on CD
(d)	Shop Test Report & Inspection Certificate	16 sets / ship
(e)	Equipment Service Log	7 sets / ship

16 **Drawing Approval.** Drawings will be approved by DSP/DQA (N). Requisite copies of drawings are to be submitted by the firm to the relevant Authorities within 4 weeks from placement of order. Approval of drawings shall be arranged by the firm.

17. **System Performance**

(a) **Reliability** . The system design should be based on standard engineering practices to provide a reliable product. The reliability figures in terms of MTBF/MTTR shall be estimated by the OEM and submitted as part of the technical proposal.

(b) **Maintainability.** The system should be able to run on line and off-line maintenance diagnostics. The minor panel along with the built in test equipment (BITE) of individual units, will be capable of detecting and localizing faults down to a single replaceable PCB/Modules within 5 minutes of the fault occurrence. The manufacturer will prepare and submit data on maintainability:-

(i) Maintainability program
(ii) Maintainability prediction

18. **Quality Assurance, Quality Control and Inspection**

(a) The Quality Assurance for software will be in accordance with IEEE STD -12207

(b) Inspection and acceptance tests will be jointly carried out by the representatives of the Indian Navy and the manufacturer. A Quality Acceptance Procedure document containing the following details will be provided to the Indian Navy.

(i) The details of Test procedure and method of proving the performance of the system both in factory and onboard ship.

(ii) Factory Acceptance Test (FAT) plan.

(iii) List of relevant hardware and software specifications to which the testing / acceptance would conform to and the performance parameters required to be checked during the various acceptance phases.

(iv) List of specifications that cannot be conclusively demonstrated /proved during System Acceptance Tests and for which documentary evidence will be furnished by the manufacturer. The manufacturer will undertake to make good, free of cost, any defects noticed during ship-borne tests/trials/operations.

19. **EMI/EMC Specifications**. The proposed system will satisfy the provisions of MIL-STD-461D or equivalent specifications for EMI/EMC requirements.

20. **ESD Protection**. The system design will take into account adequate measures for Electro Static Discharge (ESD) control and protection at PCB/module/assembly and unit level. Each Electro Static Discharge sensitive part/ assembly will be duly marked with a symbol /warning.

21. **MAINTENANCE.**

(a) **Onboard Repair / Maintenance**. Onboard repair shall be restricted to replacement at PCB level. Routine maintenance and serviceability check/ terminal performance checks would also be undertaken by ship staff as part of 1st line maintenance

(b) **Second level repair / maintenance**. Second level repair / maintenance of the system would normally be undertaken by shore based unit (dockyard) and will be resorted to when:-

(i) Second level maintenance involving higher skills and /or elaborate set ups are required.

(ii) Periodic parameter checks and record of performance is to be undertaken as part of STW/HATs or after major components / sub-units of the system have been replaced.

(c) **Third/Fourth Level Repair / Maintenance**. Third/fourth level repair / maintenance would include following:-

(i) PCB /repairable module repairs down to component level.

(ii) Repair/overhaul/refurbishing and testing of major assemblies/complete system.

(iii) Setting to work, testing and tuning onboard ships after major repairs /overhauls.

(iv) Repair and calibration of BITE and other test equipment.

22. **Equipment Spares**

(a) **Manufacturers' recommended spares**. The manufacturer will prepare the list of spares taking into account the IN's maintenance philosophy and test equipment policy. The basis for ranging and scaling of the spares will be clearly indicated. Unit cost with Pattern numbers of each item is required to be specified. The part Identification List should conform to the Integrated Logistic Management System format of the Indian Navy. The lead-time for the supply of items is also be specified. The following are the requirements of the Indian Navy in so far as the spares are concerned.

(i) Provision of On Board spares based on two year's requirement with a maximum mission time of 90 days.

(ii) Provision of depot spares based on 5 years stock required for carrying out the functions listed at para 15(c) above, to meet any contingencies and replenish the onboard spares for supporting systems for 5 years.

(iii) Scope, modalities and pricing for long term supply of spares ensuring the life of the system and undertake to maintain the supply of same for at least a period of 12 years.

(iv) Provide working substitute for those components becoming obsolete for a period of 2 years.

(b) **Test Software & Test Procedures For PCB/Modules Repair.** The manufacturer will include the following in the project proposal:-

(i) Test software compatible to automatic PCB testers available with Indian Navy.

(ii) Additional test jigs/facilities for fault localization/repair of hybrid modules/analog/high density PCB's, together with test procedures, waveforms etc.

(iii) Cost in respect of the above

23. **Weight and dimensions.** Details are to be intimated as part of the documentation.

24. **Warranty.** This will be for a period of 18 months from the commissioning and proving of the equipment onboard.

25 **Test Equipment**. The OEM to provide details of following test equipment that would be required.

(a) **Onboard test equipment.** This should be adequate to meet the requirement of all repairs/maintenance expected to be carried out onboard the ship by the ship staff.

(b) **Special test equipment.** Special to – type test equipment shall be made available in the shore base for specific tests/ CHECKS ON THE EQUIPMENT.

(C) **Dockyard test equipment.** This should cater for the functions listed at para 15(c) above.

26. **Comprehensive Maintenance Contract** : The firm should be willing to provide CMC support for a duration post warranty, which could be finalized at the time of contract.